2.25 Differentiation: The Quotient Rule

Worked Solutions 1–34

1 a)

$f'(x) = \frac{3(2x-1)-(3x+1)2}{(2x-1)^2} = \frac{6x-3-6x-2}{(2x-1)^2} = -\frac{5}{(2x-1)^2}$.

1 b)

$f'(x) = \frac{1(1-4x)-x(-4)}{(1-4x)^2} = \frac{1-4x+4x}{(1-4x)^2} = \frac{1}{(1-4x)^2}$.

1 c)

$f'(x) = \frac{3(3x-1)-(3x-3)(3)}{(3x-1)^2} = \frac{6x-3-6x+9}{(3x-1)^2} = \frac{6}{(3x-1)^2}$.

1 d)

$f'(x) = \frac{e^x(x-3)-e^x(1)}{(x-3)^2} = \frac{xe^x-3e^x-e^x}{(x-3)^2} = \frac{e^x(x-4)}{(x-3)^2}$.

1 e)

$f'(x) = \frac{\frac{1}{x}x^3-3x^2\ln x}{x^6} = \frac{x^2-3x^2\ln x}{x^6} = \frac{1-3\ln x}{x^4}$.

1 f)

$$f'(x) = \frac{10\cos(10x)\times 2x^3 - \sin(10x)\times 6x^2}{(2x^3)^2}$$
$$= \frac{2x^2(10x\cos(10x) - 3\sin(10x))}{4x^6} = \frac{10x\cos(10x) - 3\sin(10x)}{2x^4}$$

1 g)

$f'(x) = \frac{\ln 5\times 5^x\times x^2-5^x\times 2x}{x^4} = \frac{5^x(x\ln 5-2)}{x^3}$.

1 h)

$f'(x) = \frac{2e^{2x}\cos x+e^{2x}\sin x}{(\cos x)^2} = \frac{e^{2x}(2\cos x+\sin x)}{\cos^2 x}$.

1 i)

$f'(x) = \frac{\frac{1}{x}(2x+1)-2\ln 5x}{(2x+1)^2} = \frac{2x+1-2x\ln 5x}{x(2x+1)^2}$.

1 j)

$f'(x) = \frac{2x\sin x-(x^2-1)\cos x}{(\sin x)^2} = \frac{2x\sin x-x^2\cos x+\cos x}{\sin^2 x}$.

2)

$$\frac{dy}{dx} = \frac{1(\sqrt{x}) - (x-5)(\frac{1}{2}x^{-\frac{1}{2}})}{x} = \frac{\sqrt{x} - \frac{1}{2}x^{\frac{1}{2}} + 5\times\frac{1}{2}x^{-\frac{1}{2}}}{x}$$
$$= \frac{\sqrt{x} - \frac{1}{2}\sqrt{x} + \frac{5}{2}\frac{1}{\sqrt{x}}}{x} = \frac{\frac{1}{2}\sqrt{x} + \frac{5}{2\sqrt{x}}}{x} = \frac{x+5}{2x\sqrt{x}}$$

3)

$$\frac{dy}{dx} = \frac{\ln 3 \times 3^x(x^2 \ln 3) - (3^x)(2x \ln 3)}{(x^2 \ln 3)^2}$$
$$= \frac{\ln 3 \times 3^x x(x \ln 3 - 2)}{x^4(\ln 3)^2} = \frac{3^x(x \ln 3 - 2)}{x^3 \ln 3}$$

4 a)

$\frac{(x+1)[(x+1)-3x]}{(x+1)^4} = \frac{(x+1)(1-2x)}{(x+1)^4} = \frac{1-2x}{(x+1)^3}$.

4 b)

$\frac{(x-4)[(x-4)-2(x+1)]}{(x-4)^4} = \frac{(x-4)[x-4-2x-2]}{(x-4)^4} = \frac{-x-6}{(x-4)^3} = -\frac{x+6}{(x-4)^3}$.

4 c)

$\frac{2(x+2)^2[(x+2)-2(7x+1)]}{(x+2)^6} = \frac{2[x+2-14x-2]}{(x+2)^4} = \frac{2(-13x)}{(x+2)^4} = -\frac{26x}{(x+2)^4}$.

4 d)

$$\frac{(x+3)^3(5x-1)^2[4(5x-1)-3(x+3)]}{(5x-1)^6}$$
$$= \frac{(x+3)^3[20x-4-3x-9]}{(5x-1)^4} = \frac{(x+3)^3(17x-13)}{(5x-1)^4}$$

4 e)

$$\frac{-6(1-2x)^3(2x+5)^2[(2x+5)-(1-2x)]}{(1-2x)^8}$$
$$= -\frac{6(2x+5)^2(4x+4)}{(1-2x)^5} = -\frac{24(2x+5)^2(x+1)}{(1-2x)^5}$$

5)

$$\frac{(x+1)e^x - \frac{1}{2}e^x}{(x+1)(x+1)^{\frac{1}{2}}} = \frac{e^x[(x+1)-\frac{1}{2}]}{(x+1)^{\frac{3}{2}}} = \frac{e^x(x+\frac{1}{2})}{(x+1)^{\frac{3}{2}}}$$
$$= \frac{e^x(2x+1)}{2(x+1)^{\frac{3}{2}}} \text{ as req}$$

6 a)

$$f'(x) = \frac{4(x+1)^2 - 4x(2(x+1))}{(x+1)^4} = \frac{4(x+1)^2 - 8x(x+1)}{(x+1)^4}$$
$$= \frac{4(x+1)[(x+1)-2x]}{(x+1)^4} = \frac{4(1-x)}{(x+1)^3}$$

6 b)

$f'(x) = \frac{9(1+2x)^3-(9x-2)\times 6(1+2x)^2}{(1+2x)^6} = \frac{3(1+2x)^2[3(1+2x)-2(9x-2)]}{(1+2x)^6} = \frac{21-36x}{(1+2x)^4}$.

6 c)

$f'(x) = \frac{4e^{4x}(x+1)^2-2e^{4x}(x+1)}{(x+1)^4} = \frac{2e^{4x}(x+1)[2(x+1)-1]}{(x+1)^4} = \frac{2e^{4x}(2x+1)}{(x+1)^3}$.

7 a)

$$\begin{aligned}\frac{dy}{dx} &= \frac{4(2x+5)^4 - 8(4x-35)(2x+5)^3}{(2x+5)^8} \\ &= \frac{4(2x+5)^3[(2x+5)-2(4x-35)]}{(2x+5)^8} \\ &= \frac{4(-6x+75)}{(2x+5)^5}\end{aligned}$$

Stat pt when $\frac{dy}{dx} = 0$. $4(-6x+75) = 0 \Rightarrow x = \frac{75}{6} = \frac{25}{2}$.

7 b)

$$\begin{aligned}\frac{dy}{dx} &= \frac{-4(3-2x)(1+x)^3 - 3(3-2x)^2(1+x)^2}{(1+x)^6} \\ &= \frac{-(3-2x)(1+x)^2[4(1+x)+3(3-2x)]}{(1+x)^6} \\ &= -\frac{(3-2x)(13-2x)}{(1+x)^4}\end{aligned}$$

Stat pt when $\frac{dy}{dx} = 0$. $-(3-2x)(13-2x) = 0 \Rightarrow x = \frac{3}{2}$ or $x = \frac{13}{2}$.

7 c)

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{1}{2}e^{\frac{1}{2}x} \times 2(x-2)^2 - e^{\frac{1}{2}x} \times 4(x-2)}{(2(x-2)^2)^2} \\ &= \frac{e^{\frac{1}{2}x}(x-2)[(x-2)-4]}{4(x-2)^4} = \frac{e^{\frac{1}{2}x}(x-6)}{4(x-2)^3}\end{aligned}$$

Stat pt when $\frac{dy}{dx} = 0$. $e^{\frac{1}{2}x}(x-6) = 0 \Rightarrow x = 6$.

7 d)

$$\begin{aligned}\frac{dy}{dx} &= \frac{ae^{ax} \times (x+a)^2 - e^{ax} \times 2(x+a)}{((x+a)^2)^2} \\ &= \frac{e^{ax}(x+a)[a(x+a)-2]}{(x+a)^4} = \frac{e^{ax}(ax+a^2-2)}{(x+a)^3}\end{aligned}$$

Stat pt when $\frac{dy}{dx} = 0$. $e^{ax}(ax+a^2-2) = 0 \Rightarrow x = \frac{2}{a} - a$.

8 a)

$$\frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$
$$= \frac{1}{\cos^2 x} = \sec^2 x$$

8 b)

$$\frac{d}{dx}\left[\frac{1}{\sin x}\right] = \frac{0 - 1 \times \cos x}{\sin^2 x} = -\frac{\cos x}{\sin x \sin x}$$
$$= -\frac{1}{\sin x} \times \frac{\cos x}{\sin x} = -\operatorname{cosec} x \cot x$$

8 c)

$$\frac{d}{dx}\left[\frac{1}{\cos x}\right] = \frac{0 - 1 \times -\sin x}{\cos^2 x} = \frac{\sin x}{\cos x \cos x}$$
$$= \frac{1}{\cos x} \times \frac{\sin x}{\cos x} = \sec x \tan x$$

9 a)

$\frac{dy}{dx} = 3\sec^2(3x)$.

9 b)

$\frac{dy}{dx} = -2\operatorname{cosec} 2x \cot 2x$.

9 c)

$\frac{dy}{dx} = \frac{1}{3}\sec\frac{1}{3}x \tan\frac{1}{3}x$.

9 d)

$\frac{dy}{dx} = -2\pi\sec^2(\pi x)$.

9 e)

$\frac{dy}{dx} = -\frac{1}{5}\sec^2(-\frac{1}{5}x)$.

9 f)

$\frac{dy}{dx} = -9\sec 9x \tan 9x$.

9 g)

$\frac{dy}{dx} = \frac{1}{2}\operatorname{cosec}(-\frac{1}{2}x)\cot(-\frac{1}{2}x) = \frac{1}{2}\operatorname{cosec}\frac{1}{2}x \cot\frac{1}{2}x$.

10)

$$\frac{dy}{dx} = \frac{2(4x^2+3)-(2x-1)(8x)}{(4x^2+3)^2} = \frac{8x^2+6-16x^2+8x}{(4x^2+3)^2}$$
$$= \frac{-8x^2+8x+6}{(4x^2+3)^2}$$

Decreasing when $\frac{dy}{dx} < 0$. $-8x^2+8x+6 < 0 \Rightarrow -2(4x^2-4x-3) < 0$. $\Rightarrow -2(2x+1)(2x-3) < 0$. So decreasing when $x < -\frac{1}{2}$, $x > \frac{3}{2}$. $x \in (-\infty, -\frac{1}{2}) \cup (\frac{3}{2}, \infty)$.

11)

$$\frac{dy}{dx} = \frac{2(4x-7)^2-(2x-4)8(4x-7)}{(4x-7)^4}$$
$$= \frac{2(4x-7)[(4x-7)-4(2x-4)]}{(4x-7)^4}$$
$$= \frac{2(9-4x)}{(4x-7)^3} = \frac{18-8x}{(4x-7)^3}$$

Stat pt when $\frac{dy}{dx} = 0$. $18-8x = 0 \Rightarrow x = \frac{9}{4}$ so $y = \frac{1}{8}$, therefore $(\frac{9}{4}, \frac{1}{8})$.

$$\frac{d^2y}{dx^2} = \frac{-8(4x-7)^3-12(18-8x)(4x-7)^2}{(4x-7)^6}$$

$\left.\frac{d^2y}{dx^2}\right|_{x=\frac{9}{4}} = -1 < 0$ so $(\frac{9}{4}, \frac{1}{8})$ is a maximum.

12 a)

$f'(x) = -\frac{\pi}{2}\sec(\pi x)\tan(\pi x)$. Stat pt when $\frac{dy}{dx} = 0$. $-\frac{\pi}{2}\sec(\pi x)\tan(\pi x) = 0$. $\Rightarrow \frac{\sin(\pi x)}{\cos^2(\pi x)} = 0 \Rightarrow \sin(\pi x) = 0 \Rightarrow \pi x = \pi$. $\Rightarrow x = 1$. So $y = -\frac{1}{2}\sec(\pi) = -\frac{1}{2\cos(\pi)}$, therefore $(1, \frac{1}{2})$.

12 b)

$f''(x) = -\frac{\pi^2}{2}\sec^3(\pi x) - \frac{\pi^2}{2}\sec(\pi x)\tan^2(\pi x)$.

$$f''(1) = -\frac{\pi^2}{2}\sec^3(\pi) - \frac{\pi^2}{2}\sec(\pi)\tan^2(\pi)$$
$$= -\frac{\pi^2}{2} \times -1 - 0 = \frac{\pi^2}{2} > 0$$

Therefore a minimum.

13)

$$\frac{dy}{dx} = \frac{\frac{1}{x} \times x^2 - \ln x \times 2x}{x^4} = \frac{x - 2x\ln x}{x^4} = \frac{1-2\ln x}{x^3}$$

Point of inflection when $\frac{d^2y}{dx^2} = 0$.

$$\frac{d^2y}{dx^2} = \frac{(-\frac{2}{x})x^3 - 3(1-2\ln x)x^2}{x^3} = \frac{-2x^2-3(1-2\ln x)x^2}{x^3}$$
$$= \frac{-2-3(1-2\ln x)}{x}$$

So $-2-3(1-2\ln x) = 0$. $\Rightarrow 1-2\ln x = -\frac{2}{3} \Rightarrow \ln x = \frac{5}{6} \Rightarrow x = e^{\frac{5}{6}}$. $\left.\frac{dy}{dx}\right|_{x=e^{\frac{5}{6}}} = \frac{1-2\ln(e^{\frac{5}{6}})}{(e^{\frac{5}{6}})^3} \neq 0$ so a single non-stationary point of inflection.

14 a)

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x^{-\frac{1}{2}}+3)(x-2)-(2\sqrt{x}+3x-9)}{(x-2)^2} \\ &= \frac{(\sqrt{x}+3x-2x^{-\frac{1}{2}}-6)-(2\sqrt{x}+3x-9)}{(x-2)^2} \\ &= \frac{-\sqrt{x}-2x^{-\frac{1}{2}}+3}{(x-2)^2} = \frac{-x-2+3\sqrt{x}}{\sqrt{x}(x-2)^2}\end{aligned}$$

Stat point when $\frac{dy}{dx} = 0$. $\Rightarrow -x-2+3\sqrt{x} = 0 \Rightarrow -x^1+3x^{\frac{1}{2}}-2 = 0$. Hidden quadratic: $\Rightarrow -(\sqrt{x}-2)(\sqrt{x}-1) = 0$. So $\sqrt{x} = 2$ or $\sqrt{x} = 1$. Stat points: $(4, \frac{7}{2})$ and $(1, 4)$. Therefore $P(1, 4)$ and $Q(4, \frac{7}{2})$.

14 b)

$m = \frac{4-\frac{7}{2}}{1-4} = -\frac{1}{6} \Rightarrow y-4 = -\frac{1}{6}(x-1)$.

14 c)

When $x = 0$, $y = \frac{25}{6}$, and $y = 0$, $x = 25$. Area of triangle: $\frac{1}{2} \times \frac{25}{6} \times 25 = \frac{625}{12}$ units^2.

15)

$f'(x) = \frac{a(x^2-a)-(ax-1)(2x)}{(x^2-a)^2}$. $f'(0) = \frac{a(0-a)-0}{(0-a)^2} = \frac{-a^2}{a^2} = -1$ as req.

16 a)

$f'(x) = \frac{0-3(2x)}{(k+x^2)^2} = \frac{-6x}{(k+x^2)^2}$. Since $x > 0$, $f'(x)$ is always < 0 so decreasing for all values of k.

16 b)

$$\begin{aligned}f''(x) &= \frac{-6(k+x^2)^2-(-6x)4x(k+x^2)}{(k+x^2)^4} \\ &= \frac{-6(k+x^2)-(-6x)4x}{(k+x^2)^3} = \frac{-6k+18x^2}{(k+x^2)^3}\end{aligned}$$

Pt of inflection is when $f''(x) = 0$. $f''(\frac{2}{\sqrt{3}}) = 0$ gives $\frac{-6k+18(\frac{2}{\sqrt{3}})^2}{(k+(\frac{2}{\sqrt{3}})^2)^3} = 0$. $\Rightarrow -6k+18(\frac{2}{\sqrt{3}})^2 = 0 \Rightarrow k = 3(\frac{2}{\sqrt{3}})^2$. $\Rightarrow k = 4$.

17 a)

$$\begin{aligned}f'(x) &= \frac{-e^{-x}(k+10x^2)-20xe^{-x}}{(k+10x^2)^2} = \frac{-e^{-x}(10x^2+20x+k)}{(k+10x^2)^2} \\ &= \frac{-(10x^2+20x+k)}{e^x(k+10x^2)^2}\end{aligned}$$

17 b)

Stat pt when $f'(x) = 0$. $\frac{-(10x^2+20x+k)}{e^x(k+10x^2)^2} = 0 \Rightarrow 10x^2 + 20x + k = 0$. At least one stat pt means at least one solution: $b^2 - 4ac = 20^2 - 4 \times 10 \times k \geq 0$. $400 - 40k \geq 0 \Rightarrow 400 \geq 40k \Rightarrow k \leq 10$.

18 a)

$\frac{dy}{dx} = 2x \sec^2(x^2 - 1)$.

18 b)

$\frac{dy}{dx} = -2e^{2x} \cot(e^{2x}) \operatorname{cosec}(e^{2x})$.

18 c)

$\frac{dy}{dx} = \cos x \tan(\sin x) \sec(\sin x)$.

19)

$$\begin{aligned}\frac{dy}{dx} &= \frac{3e^{3x-7}(2x-3)^2 - 4e^{3x-7}(2x-3)}{(2x-3)^4} \\ &= \frac{3e^{3x-7}(2x-3) - 4e^{3x-7}}{(2x-3)^3} = \frac{e^{3x-7}[3(2x-3)-4]}{(2x-3)^3} \\ &= \frac{e^{3x-7}(6x-13)}{(2x-3)^3}\end{aligned}$$

Stat pt when $\frac{dy}{dx} = 0$ so $e^{3x-7}(6x - 13) = 0$. Which only has one solution, $x = \frac{13}{6}$ as $e^{3x-7} \neq 0$.

20 a)

$k = \frac{1}{2}$.

20 b)

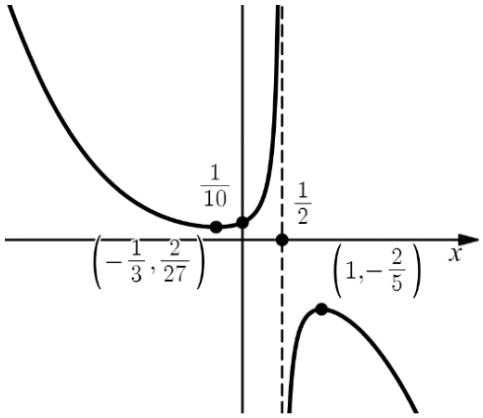
$$\begin{aligned}f'(x) &= \frac{40x(x^2+1)(1-2x) + 20(x^2+1)^2}{100(1-2x)^2} \\ &= \frac{2x(x^2+1)(1-2x) + (x^2+1)^2}{5(1-2x)^2} \\ &= \frac{2(x^2+1)(2x(1-2x) + (x^2+1))}{5(1-2x)^2} \\ &= \frac{2(x^2+1)(2x - 4x^2 + x^2 + 1)}{100(1-2x)^2} = \frac{-2(x^2+1)(3x^2-2x-1)}{100(1-2x)^2}\end{aligned}$$

Stat pt when $f'(x) = 0$. $-2(x^2 + 1)(3x^2 - 2x - 1) = 0$. So since $x^2 + 1 \neq 0 \Rightarrow 3x^2 - 2x - 1 = 0$.

20 c)

$3x^2 - 2x - 1 = 0 \Rightarrow x = -\frac{1}{3}$ or $x = 1$. Giving $(-\frac{1}{3}, \frac{2}{27})$ and $(1, -\frac{2}{5})$.

20 d)



21 a)

$\frac{dy}{dx} = \frac{2xe^{x^2}(x^2-1)-2xe^{x^2}}{(x^2-1)^2} = \frac{2xe^{x^2}(x^2-2)}{(x^2-1)^2}$.

21 b)

$$\begin{aligned}\frac{dy}{dx} &= \frac{3x^2\cos(x^3)e^{2x+1} - 2\sin(x^3)e^{2x+1}}{(e^{2x+1})^2} \\ &= \frac{3x^2\cos(x^3) - 2\sin(x^3)}{e^{2x+1}}\end{aligned}$$

21 c)

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\sin x(\ln 3)\times 3^{\cos x}\times x^2 - 2x\times 3^{\cos x}}{x^4} \\ &= -\frac{3^{\cos x}[x(\ln 3)\sin x + 2]}{x^3}\end{aligned}$$

22 a)

$\frac{dy}{dx} = 4(\frac{x^2-1}{x-3})^3 \times \frac{2x(x-3)-(x^2-1)}{(x-3)^2}$.

22 b)

$\left.\frac{dy}{dx}\right|_{x=2} = 756$ and when $x = 2$, $y = 81$. $y - 81 = 756(x-2)$. $\Rightarrow y - 81 = 756x - 1512$. $\Rightarrow y - 756x + 1431 = 0$ as req.

23 a)

$\frac{dy}{dx} = 2(\frac{\ln x}{x^2}) \times \frac{\frac{1}{x}\times x^2 - 2x\ln x}{x^4} = \frac{2\ln x - 4(\ln x)^2}{x^5}$.

23 b)

$\left.\frac{dy}{dx}\right|_{x=1} = \frac{2\ln 1 - 4(\ln 1)^2}{1^5} = 0$, therefore a stat point at $x = 1$.

24 a)

$$f'(x) = \frac{15\cos 3x \times e^{\sqrt{3}x+2} - 5\sqrt{3}\sin 3x \times e^{\sqrt{3}x+2}}{(e^{\sqrt{3}x+2})^2}$$
$$= \frac{5e^{\sqrt{3}x+2}(3\cos 3x - \sqrt{3}\sin 3x)}{(e^{\sqrt{3}x+2})^2}$$

Stat pt when $f'(x) = 0$ so when $5e^{\sqrt{3}x+2}(3\cos 3x - \sqrt{3}\sin 3x) = 0$. $\Rightarrow 3\cos 3x - \sqrt{3}\sin 3x = 0$. $\Rightarrow 3\cos 3x = \sqrt{3}\sin 3x$. $\Rightarrow 3 = \sqrt{3}\tan 3x \Rightarrow \tan 3x = \frac{3}{\sqrt{3}} = \sqrt{3}$ as req.

24 b)

$\tan 3x = \sqrt{3} \Rightarrow 3x = \frac{\pi}{3}$ or $3x = \frac{4\pi}{3}$. $\Rightarrow x = \frac{\pi}{9}$ or $x = \frac{4\pi}{9}$.

25 a)

$$f'(x) = \frac{\frac{1}{x+1}(x+1)^2 - 2(x+1)\ln(x+1)}{(x+1)^4}$$
$$= \frac{\frac{1}{x+1}(x+1) - 2\ln(x+1)}{(x+1)^3} = \frac{1-2\ln(x+1)}{(x+1)^3} \text{ as req}$$

25 b)

The point Q is when $\frac{1-2\ln(x+1)}{(x+1)^3} = 0$. $\Rightarrow 1 - 2\ln(x+1) = 0$. $\Rightarrow \ln(x+1) = \frac{1}{2} \Rightarrow x+1 = e^{\frac{1}{2}} \Rightarrow x = e^{\frac{1}{2}} - 1$.

$$\int_0^{e^{\frac{1}{2}}-1} y\,dx = \left(\frac{\ln(e^{\frac{1}{2}} - 1 + 1)}{(e^{\frac{1}{2}} - 1 + 1)^2}\right) - \left(\frac{\ln(0+1)}{(0+1)^2}\right)$$
$$= \frac{1}{2e} - 0 = \frac{1}{2e} \text{ as req}$$

26)

$$f'(x) = (1-2x)(4x+a) - 2x(2x^2 + ax + b)$$
$$f'(1) = -(4+a) - 2(2+a+b)$$
$$= -4 - a - 4 - 2a - 2b = -8 - 3a - 2b$$

$g'(x) = \frac{9a(x^2+2) - 18x(ax+b)}{(x^2+2)^2}$. $g'(1) = \frac{27a - 18(a+b)}{9} = 3a - 2(a+b) = a - 2b$. $-8 - 3a - 2b = -8$ and $a - 2b = -8$. So solve $-3a - 2b = 0$ and $a - 2b = -8$. $a = -2$, $b = 3$.

27 a)

$$f'(x) = \frac{(x^2e^x + 2xe^x)(2x-1) - 2x^2e^x}{(2x-1)^2}$$
$$= \frac{xe^x(x+2)(2x-1) - 2x^2e^x}{(2x-1)^2} = \frac{xe^x((x+2)(2x-1) - 2x)}{(2x-1)^2}$$
$$= \frac{xe^x(2x^2 + x - 2)}{(2x-1)^2}$$

27 b)

$f'(x) = \frac{(2^x \cos x + \ln 2 \times 2^x \sin x)(\sqrt{x}) - 2^x \sin x \times \frac{1}{2} x^{-\frac{1}{2}}}{x}$.

27 c)

$f'(x) = \frac{(1+\ln x)(x-1)^2 - 2x(x-1)\ln x}{(x-1)^4}$.

28)

$$\frac{dy}{dx} = \frac{((x^2-2x-3)e^x + (2x-2)e^x)(2x+1)^{\frac{1}{2}} - ((x^2-2x-3)e^x)(2x+1)^{-\frac{1}{2}}}{2x+1}$$

$\left.\frac{dy}{dx}\right|_{x=0} = -2$ when $x = 0$, $y = -3$. $y + 3 = -2x \Rightarrow y = -2x - 3$ as req.

29 a)

$$\frac{dy}{dx} = \frac{\frac{1}{2}e^{\frac{1}{2}x}(x^2+2x+1) - (2x+2)e^{\frac{1}{2}x}}{(x^2+2x+1)^2} = \frac{\frac{1}{2}e^{\frac{1}{2}x}(x^2-2x-3)}{(x^2+2x+1)^2}$$
$$= \frac{\frac{1}{2}e^{\frac{1}{2}x}(x^2-2x-3)}{(x^2+2x+1)^2} = \frac{\frac{1}{2}e^{\frac{1}{2}x}(x+1)(x-3)}{(x+1)^4} = \frac{\frac{1}{2}e^{\frac{1}{2}x}(x-3)}{(x+1)^3}$$

Stat pt when $\frac{dy}{dx} = 0$ so $\frac{\frac{1}{2}e^{\frac{1}{2}x}(x-3)}{(x+1)^3} = 0 \Rightarrow x = 3$.

29 b)

$$\frac{d^2y}{dx^2} = \frac{[\frac{1}{2}e^{\frac{1}{2}x} + \frac{1}{4}e^{\frac{1}{2}x}(x-3)](x+1)^3 - \frac{3}{2}e^{\frac{1}{2}x}(x-3)(x+1)^2}{(x+1)^6}$$
$$= \frac{\frac{1}{2}e^{\frac{1}{2}x}(x+1) + \frac{1}{4}e^{\frac{1}{2}x}(x-3)(x+1) - \frac{3}{2}e^{\frac{1}{2}x}(x-3)}{(x+1)^4}$$
$$= \frac{\frac{1}{4}e^{\frac{1}{2}x}[2(x+1) + (x-3)(x+1) - 6(x-3)]}{(x+1)^4}$$
$$= \frac{\frac{1}{4}e^{\frac{1}{2}x}[2x+2+x^2-2x-3-6x+18]}{(x+1)^4}$$
$$= \frac{e^{\frac{1}{2}x}(x^2-6x+17)}{4(x+1)^4} \text{ as req}$$

29 c)

Pts of inflection when $\frac{d^2y}{dx^2} = 0$. $\frac{e^{\frac{1}{2}x}(x^2-6x+17)}{4(x+1)^4} = 0$ as $e^{\frac{1}{2}x} \neq 0$, $x^2-6x+17 = 0$. But $(-6)^2 - 4\times1\times17 = 36 - 68 = -32 < 0$ so no real solutions.

30)

$$\frac{2\cos 2x\cos x+\sin 2x\sin x}{\cos^2 x}=\frac{2\cos 2x\cos x+2\cos x\sin^2 x}{\cos^2 x}$$
$$=\frac{2\cos 2x+2\sin^2 x}{\cos x}=\frac{2(1-2\sin^2 x)+2\sin^2 x}{\cos x}=\frac{2-2\sin^2 x}{\cos x}$$
$$=\frac{2(1-\sin^2 x)}{\cos x}=\frac{2\cos^2 x}{\cos x}=2\cos x$$

31)

$$\frac{2\cos x(5\sin x+5\cos x)-2\sin x(5\cos x-5\sin x)}{25(\sin x+\cos x)(\sin x+\cos x)}$$
$$=\frac{10\sin x\cos x+10\cos^2 x-10\sin x\cos x+10\sin^2 x}{25(\sin^2 x+2\sin x\cos x+\cos^2 x)}$$
$$=\frac{10\cos^2 x+10\sin^2 x}{5(1+2\sin x\cos x)}=\frac{10(\cos^2 x+\sin^2 x)}{25(1+2\sin x\cos x)}$$
$$=\frac{2}{5(1+\sin 2x)}$$

So $p=2$, $q=5$.

32 a)

$k=3$.

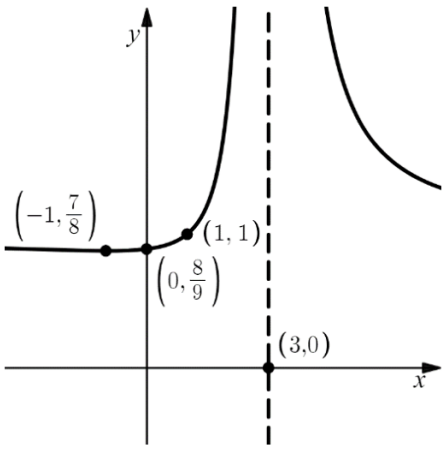
32 b)

When $x=0$, $y=-\frac{1}{9}$ so $B(0,-\frac{1}{9})$. When $y=0$, $x=1$ so $C(1,0)$.

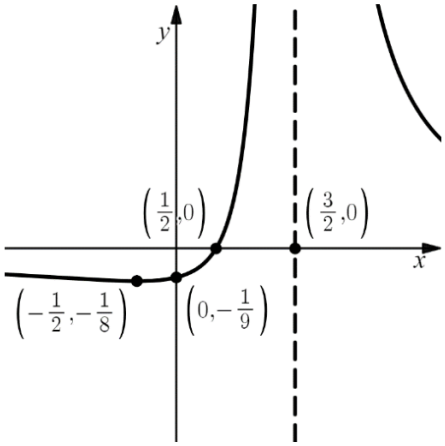
32 c)

$f'(x)=\frac{(x-3)^2-2(x-1)(x-3)}{(x-3)^4}=\frac{-x-1}{(x-3)^3}$. Stat pt when $f'(x)=0\Rightarrow -x-1=0$. $\Rightarrow x=-1$, $y=-\frac{1}{8}$ so $A(-1,-\frac{1}{8})$.

32 d i)



32 d ii)



33 a)

$\ln x - 1 = 0 \Rightarrow x = e$. And $\ln x$ gives $x > 0$. So maximum domain is $x > 0$ and $x \neq e$.

33 b)

$$f'(x) = \frac{\frac{2}{x}(\ln x - 1) - \frac{1}{x}(2\ln x - 5)}{(\ln x - 1)^2} = \frac{2(\ln x - 1) - (2\ln x - 5)}{x(\ln x - 1)^2}$$
$$= \frac{2\ln x - 2 - 2\ln x + 5}{x(\ln x - 1)^2} = \frac{3}{x(\ln x - 1)^2}$$

So since $x \neq e$, $(\ln x - 1)^2$ is always positive. Since $x > 0$, $f'(x) > 0$ so $f(x)$ is always increasing.

33 c)

It crosses the x-axis when $\frac{2\ln x - 5}{\ln x - 1} = 0$ which is when $2\ln x - 5 = 0 \Rightarrow \ln x = \frac{5}{2}$. $\Rightarrow x = e^{\frac{5}{2}}$. The function is always increasing. Therefore $f(x) < 0$ when $e < x < e^{\frac{5}{2}}$.

34 a)

$\{x : x \in \mathbb{R},\ x \neq 1\}$.

34 b)

$$f'(x) = \frac{\ln 2 \times 2^x(x-1) - 2^x}{(x-1)^2} = \frac{2^x(\ln 2(x-1) - 1)}{(x-1)^2}$$
$$= \frac{2^x(x\ln 2 - \ln 2 - 1)}{(x-1)^2}$$

Stat pt when $f'(x) = 0$ so $\frac{2^x(x\ln 2 - \ln 2 - 1)}{(x-1)^2} = 0$. $2^x \neq 0$ so $x\ln 2 - \ln 2 - 1 = 0$. $\Rightarrow x\ln 2 = \ln 2 + 1 \Rightarrow$ $x = 1 + \frac{1}{\ln 2}$. $y = 3.76833877\ldots$ so $(2.44, 3.77)$.

34 c)

$f(x) > 3.77$. But since $2^x > 0$ and $\frac{2^x}{x-1} < 0$ for $x < 1$ so $f(x) < 0$. Therefore range is:

$$\{f(x) : f(x) \in \mathbb{R},\ f(x) < 0\} \cup \{f(x) : f(x) \in \mathbb{R},\ f(x) > 3.77\}$$